Final Project Report: EV Value & Longevity Analytics Dashboard

Price-per-Range, Battery Health and Demand Insights

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1. Introduction

1.1 Project Overview

This project builds an interactive Tableau dashboard on a global EV market dataset of 200,000 vehicles from 200 manufacturers, covering model years 2018 to 2026. It adds a Value Index (price per km of range) and a battery longevity view to the usual sales and price charts.

1.2 Purpose

To give fleet buyers and pricing analysts a single place to compare what EVs cost, how far they go, how long their batteries last and how safe they are.

2. Ideation Phase

2.1 Problem Statement

Decision-makers cannot easily compare EV value across 15 categories and 200 manufacturers, because the data is in raw tables and no common value measure exists.

2.2 Empathy Map Summary

Target user is a Fleet Procurement Manager. Pains: fear of overpaying and of early battery degradation. Gains: one clear view of value, durability and safety.

2.3 Brainstorming

Seven ideas were ranked. Six were selected: the 9-chart dashboard, Value Index, battery comparison, manufacturer leaderboard, 5-chapter story and country/region map. Sales forecasting moved to future scope.

3. Requirement Analysis

3.1 Customer Journey

Four stages: land and orient, compare segments, check battery and value, then decide and share.

3.2 Solution Requirements

Functional: load and validate the CSV files, compute averages and the Value Index, build nine worksheets, add filters, publish. Non-functional: readable layout, responsive filtering, public data only, always-on web access.

3.3 Data Flow

Train and test CSV > validation and cleaning > Tableau extract > nine worksheets > dashboard and story > Tableau Public > end user.

3.4 Technology Stack

Component	Technology	Purpose
BI tool	Tableau Desktop (Public edition)	Build worksheets and dashboard
Hosting	Tableau Public	Share the interactive dashboard
Data source	CSV files (train 200,000 rows, test 100,000 rows)	Source of all measures
Validation	Python (pandas)	Check nulls, duplicates, totals

4. Project Design

Problem-solution fit: the Value Index and battery view turn raw columns into decision measures. Architecture: two CSV files feed a Tableau extract, then nine worksheets, then one dashboard and one story published to Tableau Public. Solution is fully described in documents 13, 18 and 19.

5. Project Planning and Scheduling

The work is organised into three 6-day sprints: Sprint-1 data and worksheets (28 Oct to 02 Nov), Sprint-2 refinement, filters and Value Index (03 Nov to 08 Nov), Sprint-3 story, testing and publishing (10 Nov to 15 Nov).

6. Functional and Performance Testing

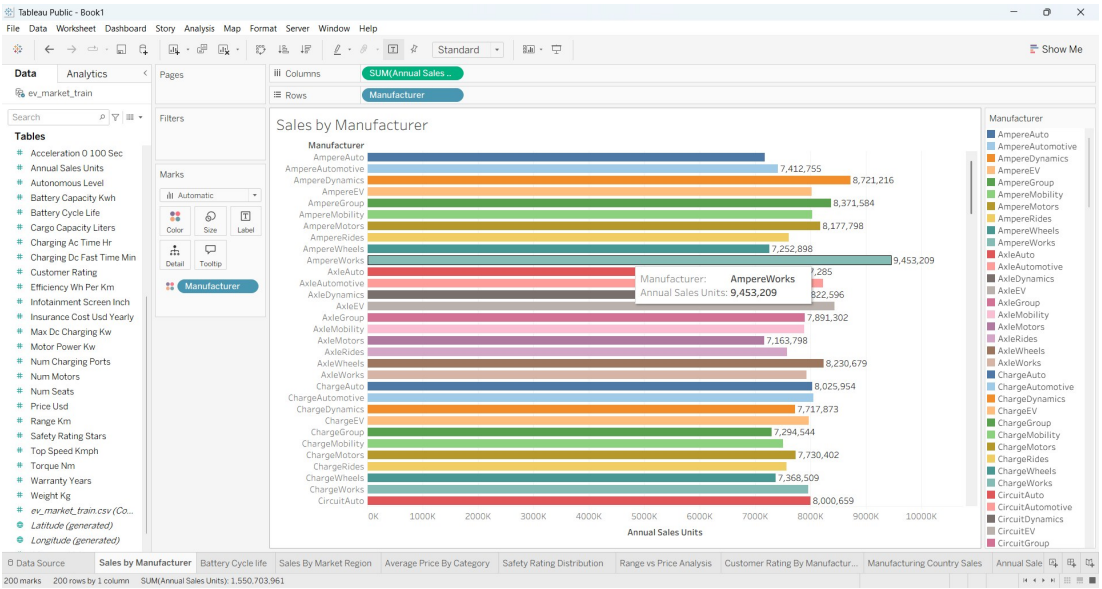
- Data rendered: 200,000 training rows and 1,550.7M annual units loaded. The 100,000-row test file has the same 37 columns and totals 772.6M units.
- Validation: test-file averages (price $65,010, range 324.3 km, rating 4.08) are within 0.2% of the training file, so the same logic holds on unseen data.
- Checks: totals in Tableau match totals computed independently in Python.
- Visuals: nine worksheets and one dashboard render correctly. Refinements are listed in document 04.

7. Results

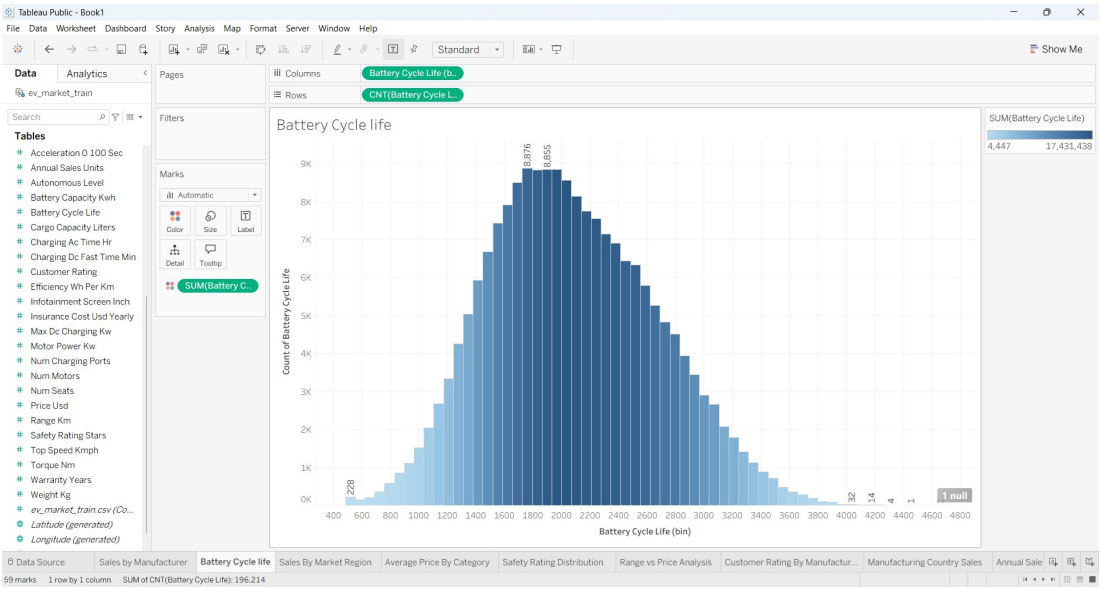
7.1 Key Findings

- Best cost per km of range: Two-Wheeler Scooter ($24 per km), Three-Wheeler ($42) and Hatchback ($82). Worst: Luxury SUV ($222) and Sports Car ($435).
- Battery life: Solid-State 2,997 cycles and 7.4% five-year degradation; LFP 2,502 and 10.1%; NMC 1,801 and 14.0%; Sodium-Ion 1,799 and 14.0%; NCA 1,603 and 14.0%.
- Safety: 76% of vehicles carry 4 or 5 stars (average 3.98).
- Price and range are related, but with diminishing returns (correlation 0.28 linear, 0.67 against log price).
- Sales are spread evenly: regions within 3%, countries within 7%, manufacturer ratings between 4.02 and 4.14.

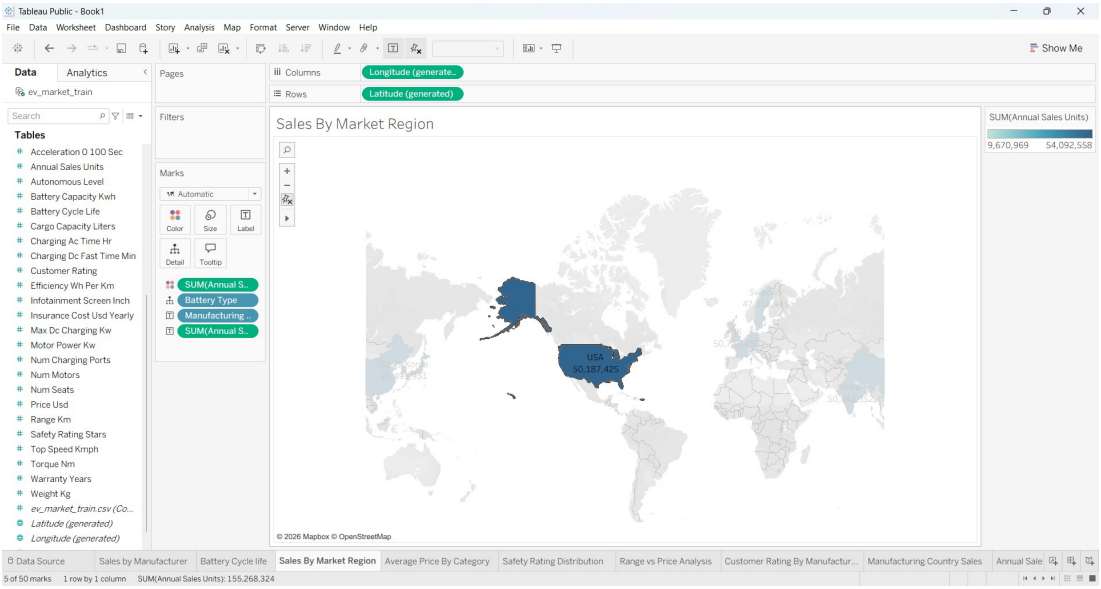
7.2 Output Screenshots



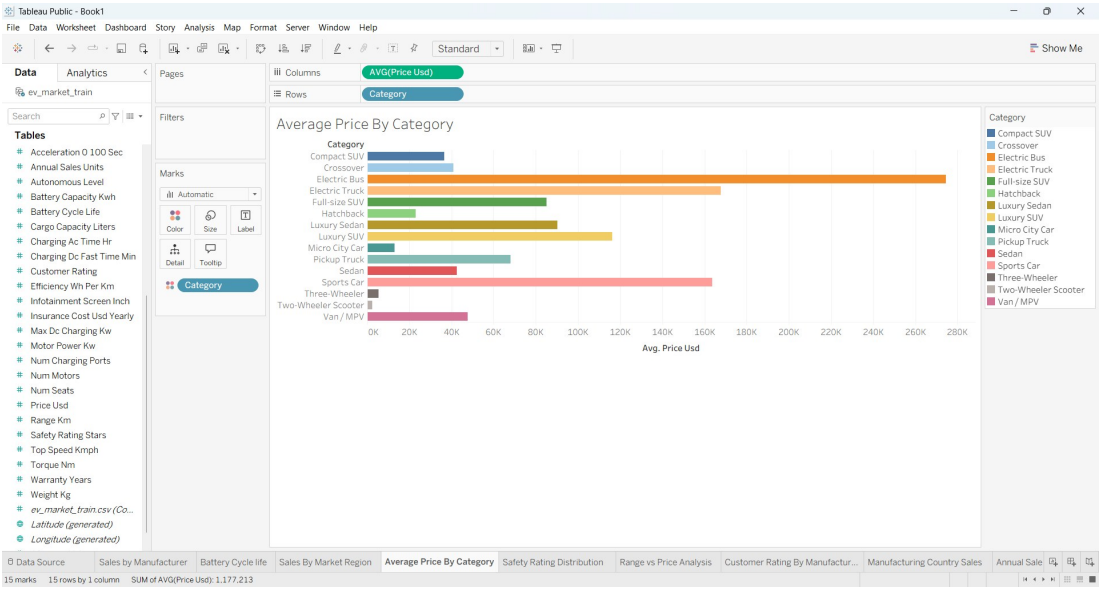
Sales by Manufacturer (Tableau worksheet)



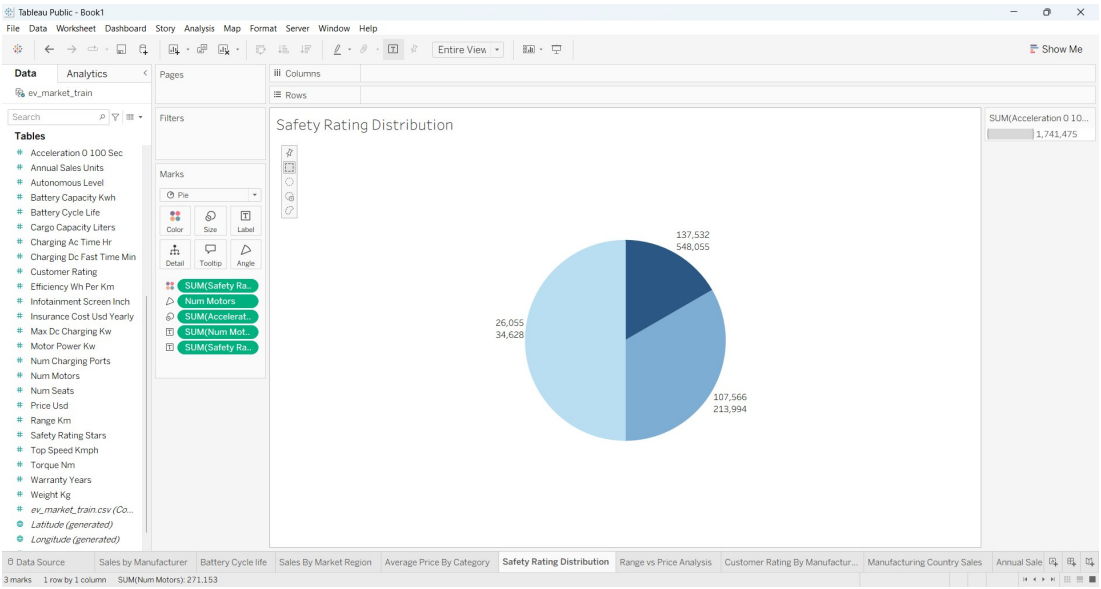
Battery Cycle Life distribution (Tableau worksheet)



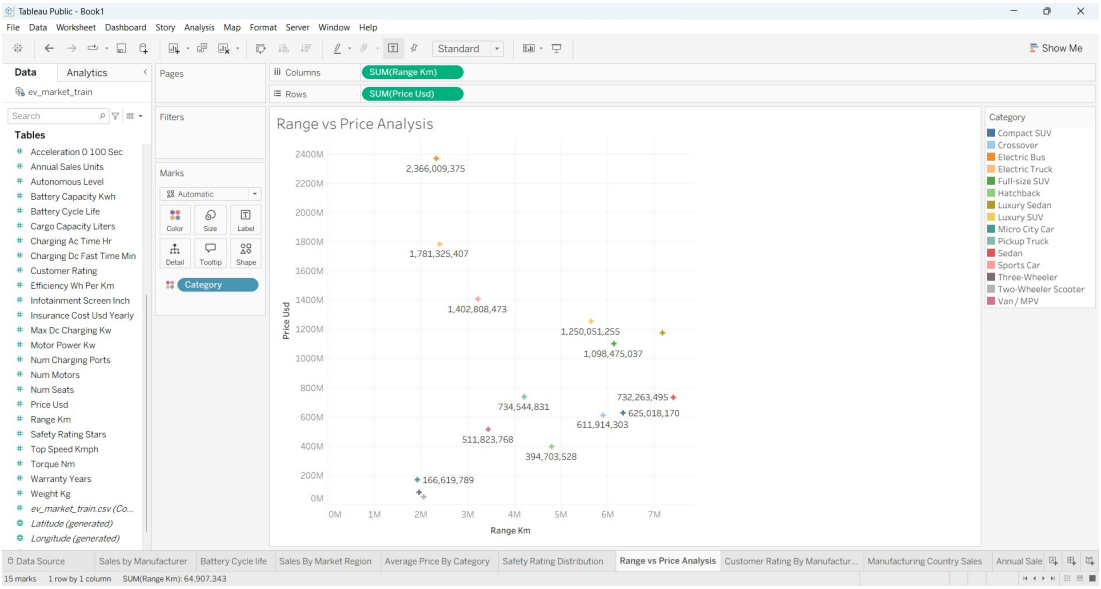
Sales by Market Region (Tableau worksheet)



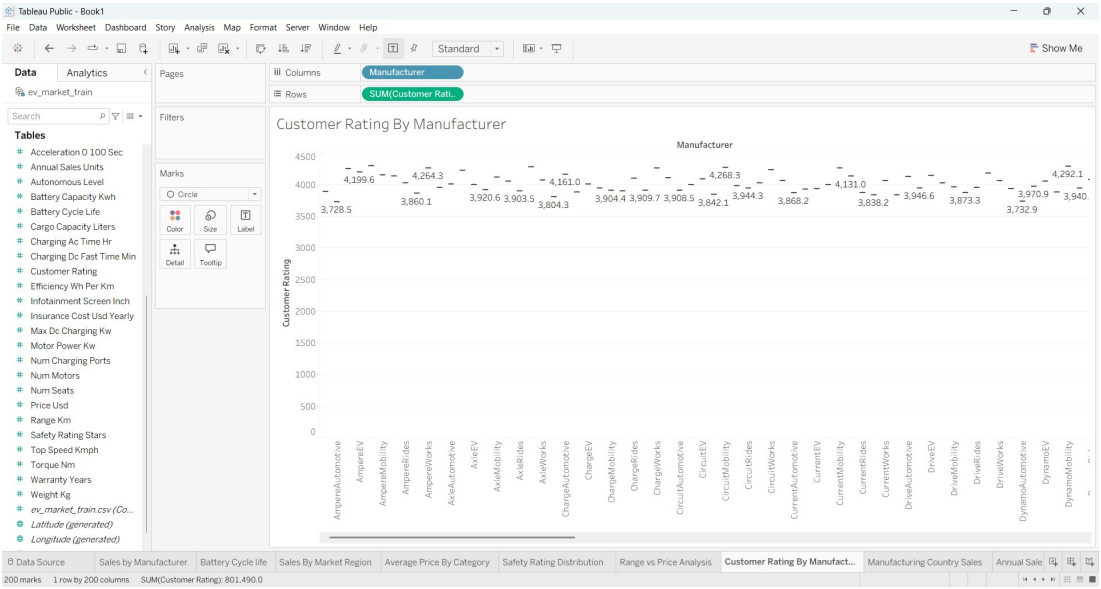
Average Price by Category (Tableau worksheet)



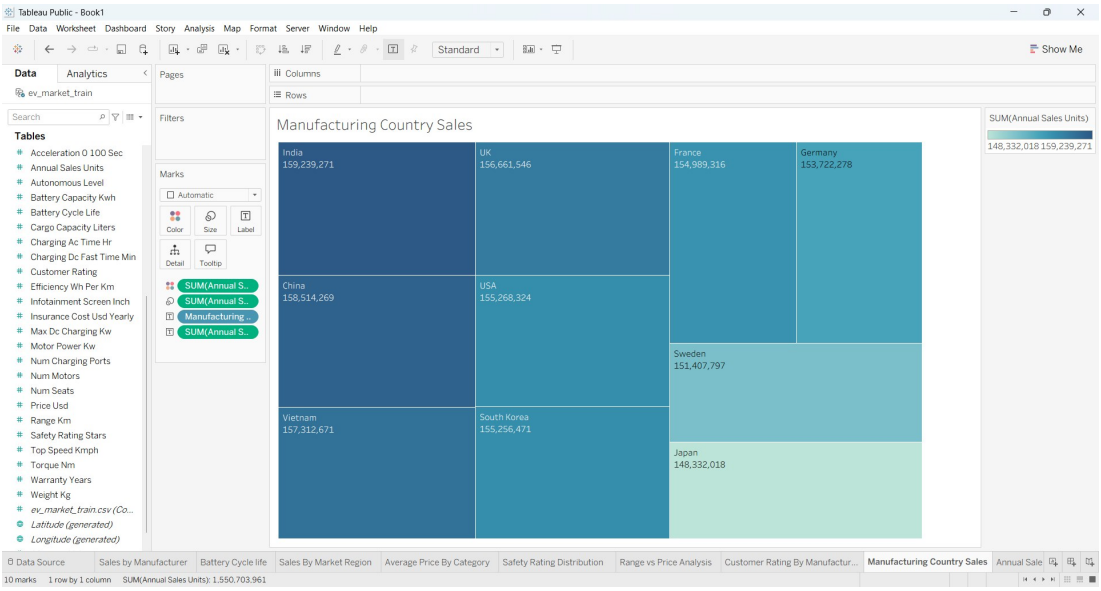
Safety Rating Distribution (Tableau worksheet)



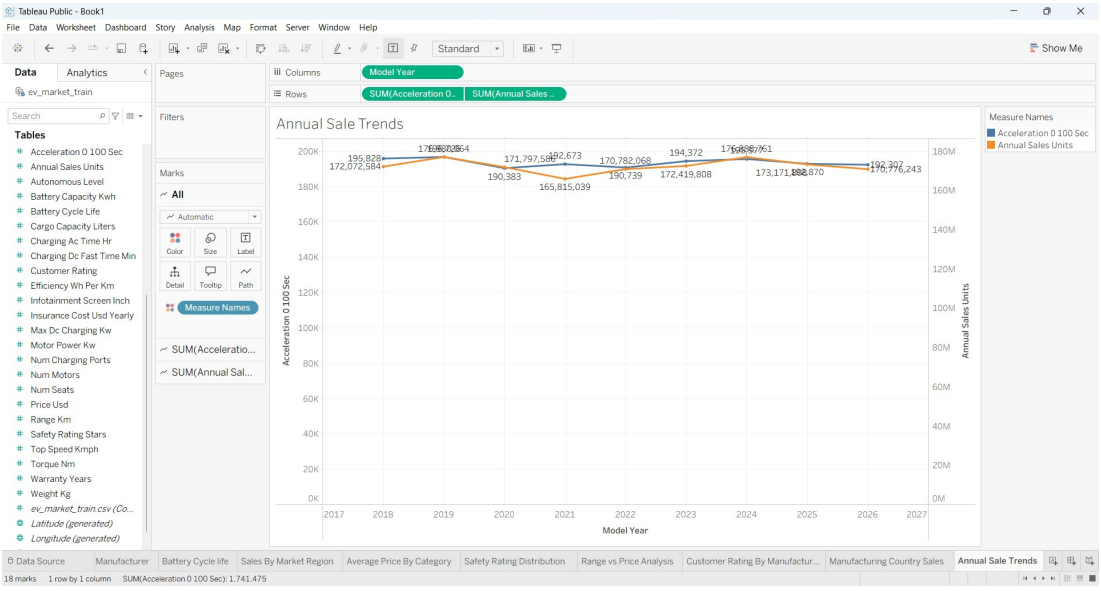
Range vs Price Analysis (Tableau worksheet)



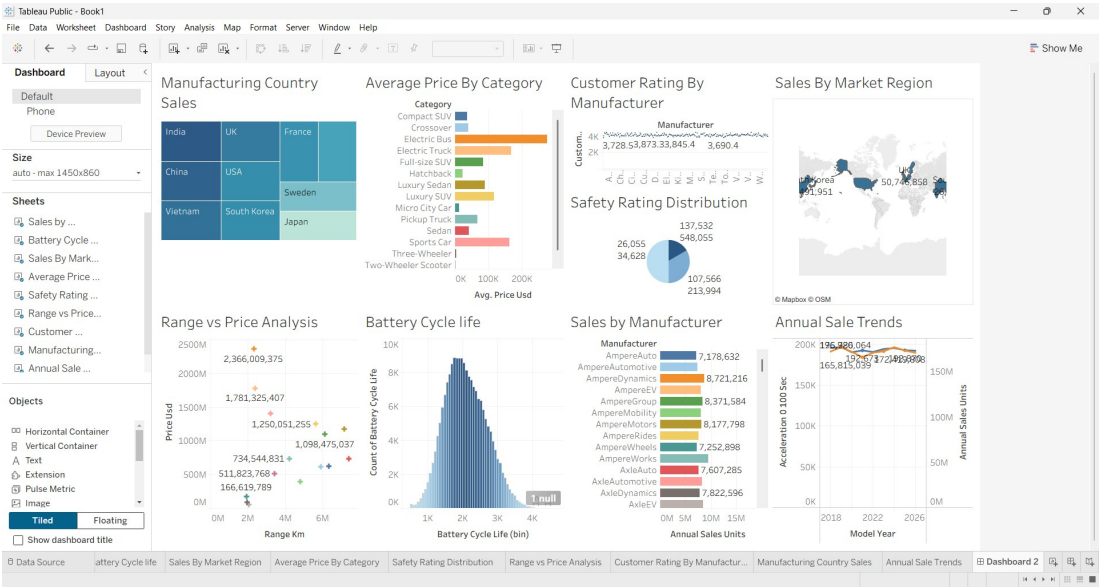
Customer Rating by Manufacturer (Tableau worksheet)



Manufacturing Country Sales (Tableau worksheet)



Annual Sale Trends (Tableau worksheet)



Final dashboard (Dashboard 2) (Tableau worksheet)

8. Advantages and Disadvantages

Advantages: one view for value, durability and safety; filters make it usable by non-technical readers; no installation needed. Disadvantages: the dataset is a static file and updates are manual; Tableau Public makes the data public; several differences between regions and ratings are small, so conclusions should be drawn carefully.

9. Conclusion

The project converts a large raw EV table into comparable measures. The clearest messages are that Solid-State and LFP batteries last longest, hatchbacks and two- and three-wheelers give the cheapest range, and luxury categories pay a steep premium for each extra km.

10. Future Scope

Train a sales forecasting model on the training file and evaluate it on the test file; connect a live database; add charging and incentive data; add a total-cost-of-ownership calculator.

11. Appendix

Dataset: ev_market_train (200,000 rows) and ev_market_test (100,000 rows), 37 columns. Story link and dashboard link: to be added after publishing on Tableau Public.